3.2.S.3.1 Elucidation of Structure and Other Characteristics [2439-exNA, Hongene]

Confirmation of the structure and characterization of 2439-exNA drug substance was accomplished by using a variety of techniques commonly used for oligonucleotides. Tests were performed on non-GMP batches of the single strands (XXXand XXX) and on the 2439-exNA drug substance GMP batch XXX. Refer to Section 3.2.S.4.4 for results obtained on drug substance batches produced.

The data provided in this section were taken from the characterization studies performed during development of 2439-exNA as well as from the Hongene batch release testing of XXX, XXX and XXX. The full set of tests performed to characterize 2439-exNA are summarized in Table 1.

Table 1. Techniques Applied to Elucidate the Structure of 2439-exNA Drug Substance

Attribute	Methods	Section
Molecular Weight	Mass spectrometry	Section 3.2.S.3.1.1
Sequence of component single strands (identity)	Sequencing by tandem mass spectrometry/mass spectrometry	Section 3.2.S.3.1.2
Extinction Coefficient	UV visible spectroscopy	Section 3.2.S.3.1.3
Sodium counterion and content	Ion chromatography	Section 3.2.S.3.1.4
Confirmation of duplex structure and Sequence (identity)	Melting temperature (T_m) by UV	Section 3.2.S.3.1.5
Appearance	Visual Inspection	Section 3.2.S.3.1.6
Identity	non-denaturing SEC	Section 3.2.S.3.1.7
Confirmation of duplex structure	Denaturing and non-denaturing UPLC	Section 3.2.S.3.1.8
Water Content	USP <921>	Section 3.2.S.3.1.9
Elucidation of structure (chromophores)	UV visible spectroscopy	Section 3.2.S.3.1.10
Physical form	-	Section 3.2.S.3.1.10

3.2.S.3.1.1 Molecular Weight by Mass Spectrometry of the Sense and Antisense Single Strands

Single strands were analyzed by mass spectrometry (ESI-MS, negative mode). The mass detector identified the negative ions of single strands with multiple charge states which are deconvoluted by an algorithm to provide the uncharged mass of the parent molecule (Figure 1 to Figure 2). The results for the two single strands are in good agreement with the theoretical values as shown in Table 2.

Table 2: Mass spectrometry of Sense and Antisense Strands of 2439-exNA

Sample	Batch No.	Theoretical value	Experimental value

sense strand	XXX	XXX	XXX
antisense strand	XXX	XXX	XXX

3.2.S.3.1.2 Molecular Sequencing of the Sense and Antisense Single Strands (MS/MS)

Both single strands (batch# XXX and XXX) were sequenced by XXX.

The fragmentation pattern of the parent ion from which the sequence is calculated from the MS/MS is provided in Figure 3 and Figure 4. Every nucleotide incorporation was confirmed by finding at least a fragment corresponding to that nucleotide.

3.2.S.3.1.3 Determination of the 2439-exNA Extinction Coefficient (ε)

The concentration for 2439-exNA is performed by UV-Visible spectrophotometer, measuring the absorbance of a solution containing the drug substance in the drug product at 260 nm after dilution in PBS for measurement. The extinction coefficient of 2439-exNA drug substance has been determined at 260 nm. The molar extinction coefficient was measured using 2439-exNA (batch number: XXX) drug substance dissolved in XXX at controlled room temperature and was determined to be XXX and when expressed as a conversion factor, XXXfor the free acid.

3.2.S.3.1.4 Sodium counterion content

2439-exNA is isolated as a sodium salt form of the conjugate base phosphodiester and phosphorothioate internucleotide linkages of the 2439-exNA duplex. Table 3 describes the calculation of the theoretical sodium content of 2439-exNA drug substance.

Table 3. 2439-exNA Molecular Weight and Formula of the Sodium Form (Salt) of 2439-exNA (Drug Substance)

Molecular Formula:	XXX
Molecular Weight:	XXX
Weight of Sodium in 2439-exNA:	XXX
Theoretical Sodium Composition:	5.95%
Measured Sodium Content:	XXX

3.2.S.3.1.5 Transition Melting Temperature of 2439-exNA Drug Substance duplex

The Tm method is used as an identification test (see Section 3.2.S.4.2) that is performed as part of routine drug substance release testing. UV absorption versus temperature profile (UV melting curve, Tm) for a duplex oligonucleotide is evaluated by UV spectrophotometer at 260 nm. DNA

or RNA duplexes show hypochromism i.e., a decrease in UV absorption upon duplex formation due to stacking (occlusion) of the heterocycles. A plot of absorbance as a function of temperature will result in a sigmoidal curve (hyperchromic shift). The inflection point (or maximum rate of change/first derivative) of this sigmoidal curve is defined as the Tm for the sample analyzed, thus confirming the presence of the duplex and the identification of the drug substance. Even a single base pair mismatch in duplexed region will have significant effects upon the stability of the duplex and thus will suppress the Tm.

The melting temperature (Tm) of 2439-exNA drug substance was measured using a XXX spectrophotometer at a concentration of XXX solution. The Tm of the 2439-exNA drug substance was XXX (see Figure 6).

A sample of 2439-exNA is solubilized and diluted to XXX in phosphate buffered saline. XXX and the absorbance of the sample at
260 nm collected. The maximum rate of change (first derivative) of the absorbance at 260 nm is the Tm.

Figure 6: Tm measurement for 2439-exNA duplex, standard and first derivative

3.2.S.3.1.6 Appearance

Visual inspection. The result of 2439-exNA drug substance (batch XXX) is XXX.

3.2.S.3.1.7 Identity (Non-denaturing SEC)

The purity of the 2439-exNA drug substance was measured by SEC under non-denaturing conditions. The retention time was consistent with the working standard (XXX). Chromatograms are reproduced in Figure 4.

3.2.S.3.1.8 Proof of duplex structure by denaturing and non-denaturing UPLC

Two chromatographic methods are used for release and retesting of the drug substance. The denaturing method (see examples in Figure 8) utilizes chaotropes, organic solvents and has a column temperature of XXXcausing denaturing of the sense and antisense strands by disrupting Watson-Crick-Franklin base pairing and the solubilizing base stacking of the nucleobases. Conversely the non-denaturing method (see example in Figure 9) is performed at lower temperature which stabilizes the duplex.

Figure 8: Denaturing AEX Chromatogram of 2439-exNA Batch #XXX

3.2.S.3.1.9 Water Content (USP<921>)

About 20 mg of 2439-exNA drug substance (XXX) was measured using the Karl-Fischer method. A water content value ofXXXwas obtained

3.2.S.3.1.10 Ultra-Violet absorption spectrum

A sample of 2439-exNA (XXX) was tested for UV absorptivity in the range XXX to XXX nm and the maximum absorptivity was determined to be XXXnm which is typical for an oligonucleotide containing a mixture of adenine, guanine, cytosine and uracil heterocycles.

3.2.S.3.1.10 Solid State Characterization

Oligonucleotide drug substances manufactured by a process involving final step lyophilization exist as amorphous powders. 2439-exNA drug substance was manufactured by final step lyophilization and, as such, exists as an amorphous powder. Given precedent within the field, no X-ray determination has been performed at this stage based on historical data.